

Developing intraosseous transcutaneous amputation prosthesis for haptic control

Author: Jonathan Holt

Supervisors: Dr Catherine Pendegrass

Dr Henry Lancashire

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Declaration

Abstract

Impact Statement

Acknowledgements

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1 Guidance

1.1 Research

Measuring devices

- strain-gauge
- accelerometer

Vibration motor

- generator for the waves (linear vs circular?) possibly build? 555 timer + amp + speaker?
- Research: vibro-tactile feedback paper. forces and frequency being used?

1.2 Writing the outline

Motivation Start with clinical problem and move in to the more focused. By the time they're done with reading my intro, they agree with why I'm making this solution

Amputations -> Loss of sensations -> How can we regain that sensation? One of the best tools is using the prosthetic itself, and we can use vibrotactile feedback

describe itap, amputations, etc ITAP as the vehicle for vibrotactile feedback. Vibrotactile feedback is one of the solutions to the problem of loss of "senses"

Risks

- Soldering risks (burning)
- Lone-working
- Long-hours

2 Outline

Aims and research question Aims are broad example improve quality of life of ALS patients.

List of specific objectives -

2.1 Motivation

Amputations as a whole

Limb loss and challenges with stumps Stump prosthesis have issues with Levy (1980):

-

ITAP as an alternative Attachment of the artificial limb directly to bone, as opposed to a soft tissue stump. What are the advantages?

ITAP as a tool for tactile feedback

- Recovery of some 'gross tact'
- Stepping-stone to cognitive control of artificial limbs. Not just controlling, but receiving tactile feedback from that control
- ITAP as the vehicle for vibrotactile feedback

2.2 The problem

Vibrations through the ITAP could be detrimental to skin-implant and bone-implant interfaces. Aim here is to investigate the magnitude and nature of the forces experienced at the interfaces. (add cause and consequence??)

2.3 Approach

3 Visualisations

1. Forces experienced at the **flange** & at the **stem**
2. Strain parallel to ITAP (Elastic modulus for titanium 6/4 is 127 GPa, so possibly infinitesimal strain theory?)

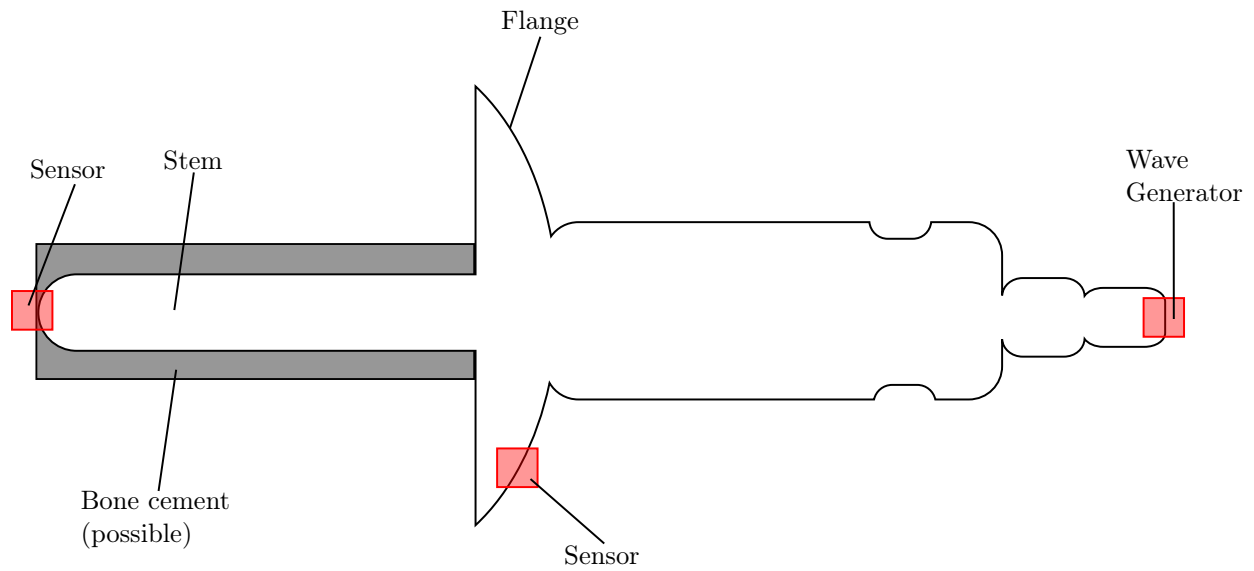
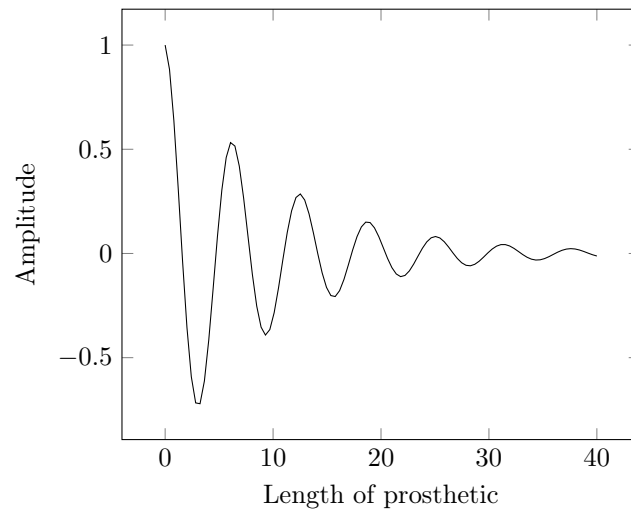


Figure 1:

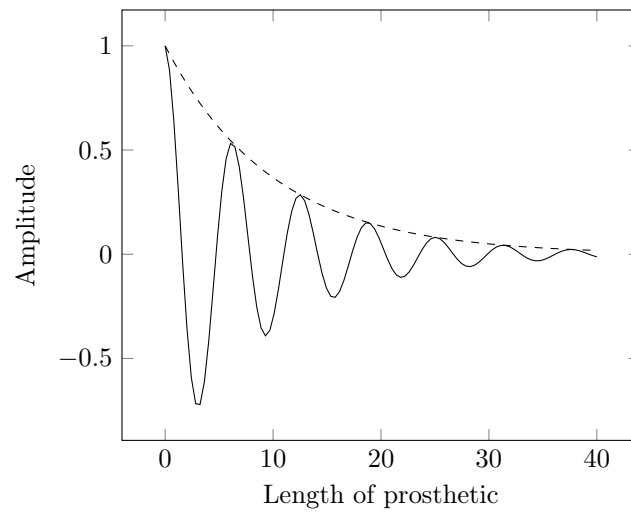
3.

1. Stem has varying lengths depending on length of bone remaining after amputation.
2. We could pick a couple of different lengths or have the assumption that after finding the dampen ratio, we can describe all sizes.
3. Characteristics of the medium change?

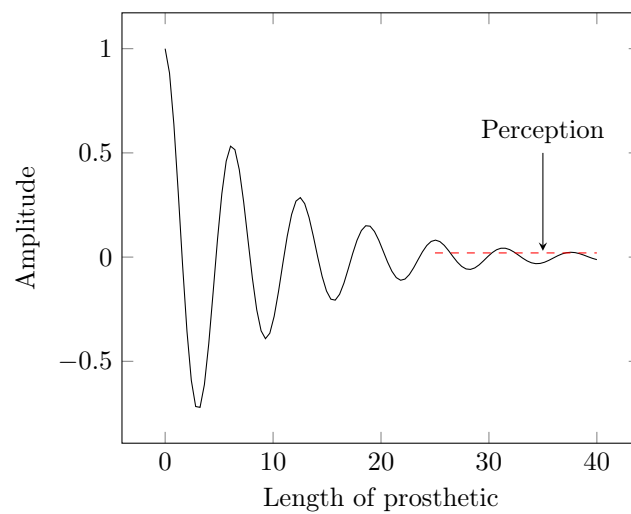
Generate wave forms



Define damping ratio



(?) Determine threshold of perception



References

- S. William Levy. Skin problems of the leg amputee. *Prosthetics and Orthotics International*, 4(1):37–44, April 1980. ISSN 0309-3646. doi: 10.3109/03093648009103113.